



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Subject: Introduction to Electronics Engineering

Sem: I 'B'

Assignment Questions-2 (Module 1 -Part A)

- 1. What is a regulated power supply? With neat block diagram explain the working of DC power supply. Also mention the principal components used in each block.**
- 2. With neat circuit diagram and waveforms explain half wave rectifier and full wave bridge rectifiers.**
- 3. Define rectifier. Explain a biphase full wave rectifier. Show the appropriate waveforms.**
- 4. Discuss the need of filter circuit. With circuit diagram and waveforms explain the operation of smoothing C – filter for half wave rectifier.**
- 5. What is voltage regulator? With neat circuit diagram, explain the operation of a voltage regulator using zener diode.**
- 6. What is voltage multiplier? With circuit diagram explain the operation of voltage doubler and voltage Tripler**
- 7. A mains transformer having a turns ratio of 22:1 is connected to a 220 V r.m.s. mains supply. If the secondary output is applied to a half-wave rectifier, determine i) secondary r.m.s voltage ii) peak voltage and iii) regulated DC voltage across load.**
- 8. A half wave rectifier uses a silicon diode is fed from a supply of 240 V, 50 Hz with step down transformer of ratio 3:1. Resistive load connected is 10 K Ω . The diode forward resistance is 75 Ω and transformer secondary is 10 Ω . Calculate i) secondary r.m.s voltage ii) peak voltage and iii) regulated DC voltage across load.**
- 9. The RC smoothing filter in a 50 Hz mains operated a half-wave rectifier circuit consists of $R_1 = 150\Omega$ and $C_2 = 1,000\mu F$. If 1 V of ripple appears at the input of the circuit, determine the amount of ripple appearing at the output.**
- 10. A 5V zener diode has a maximum rated power dissipation of 500 mW. If the diode is to be used in a simple regulator circuit to supply a regulated 5V to a load having a resistance of 500 Ω , determine a suitable value of series resistor for operation in conjunction with a supply of 9V.**